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Direct Preference Optimization: Your Language Model is Secretly a Reward Model

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ABSTRACT

While large-scale unsupervised language models (LMs) learn broad world knowledge and some reasoning skills, achieving precise control of their behavior is difficult due to the completely unsupervised nature of their training. Existing methods for gaining such steerability collect human labels of the relative quality of model generations and fine-tune the unsupervised LM to align with these preferences, often with reinforcement learning from human feedback (RLHF). However, RLHF is a complex and often unstable procedure, first fitting a reward model that reflects the human preferences, and then fine-tuning the large unsupervised LM using reinforcement learning to maximize this estimated reward without drifting too far from the original model. In this paper we introduce a new parameterization of the reward model in RLHF that enables extraction of the corresponding optimal policy in closed form, allowing us to solve the standard RLHF problem with only a simple classification loss. The resulting algorithm, which we call *Direct Preference Optimization* (DPO), is stable, performant, and computationally lightweight, eliminating the need for sampling from the LM during fine-tuning or performing significant hyperparameter tuning. Our experiments show that DPO can fine-tune LMs to align with human preferences as well as or better than existing methods. Notably, fine-tuning with DPO exceeds PPO-based RLHF in ability to control sentiment of generations, and matches or improves response quality in summarization and single-turn dialogue while being substantially simpler to implement and train.

Abridged from Rafailov, R. et al. (2023), Advances in Neural Information Processing Systems 36, arXiv:2305.18290v3, CC BY 4.0 · equations renumbered, Figure 1 redrawn

1 Introduction

Large unsupervised language models (LMs) trained on very large datasets acquire surprising capabilities (Brown *et al.*, 2020; Chowdhery *et al.*, 2022; Bubeck *et al.*, 2023; Touvron *et al.*, 2023). However, these models are trained on data generated by humans with a wide variety of goals, priorities, and skillsets. In other words, selecting the model's *desired responses and behavior* from its very wide *knowledge and abilities* is crucial to building AI systems that are safe, performant, and controllable (Ouyang *et al.*, 2022). While existing methods typically steer LMs to match human preferences using reinforcement learning (RL), we will show that the RL-based objective used by existing methods can be optimized exactly with a simple binary cross-entropy objective, greatly simplifying the preference learning pipeline.

In this paper, we show how to directly optimize a language model to adhere to human preferences, without explicit reward modeling or reinforcement learning. We propose *Direct Preference Optimization (DPO)*, an algorithm that implicitly optimizes the same objective as existing RLHF algorithms (reward maximization with a KL-divergence constraint) but is simple to implement and straightforward to train. Intuitively, the DPO update increases the relative log probability of preferred to dispreferred responses, but it incorporates a dynamic, per-example importance weight that prevents the model degeneration that we find occurs with a naive probability ratio objective. Like existing algorithms, DPO relies on a theoretical preference model, such as the Bradley-Terry model of Bradley and Terry (1952), that measures how well a given reward function aligns with empirical preference data. However, while existing methods use the preference model to define a preference loss to train a reward model and then train a policy that optimizes the learned reward model, DPO uses a change of variables to define the preference loss as a function of the policy directly. Given a dataset of human preferences over model responses, DPO can therefore optimize a policy using a simple binary cross entropy objective, producing the optimal policy to an implicit reward function fit to the preference data.

Our main contribution is Direct Preference Optimization (DPO), a simple RL-free algorithm for training language models from preferences. Our experiments show that DPO is at least as effective as existing methods, including PPO-based RLHF, for learning from preferences in tasks such as sentiment modulation, summarization, and dialogue, using language models with up to 6B parameters.

3 Preliminaries

We review the RLHF pipeline in Ziegler et al. (2020) (Bai *et al.*, 2022; Ouyang *et al.*, 2022; and later Stiennon *et al.*, 2022). It usually includes three phases: 1) supervised fine-tuning (SFT); 2) preference sampling and reward learning and 3) RL optimization.

SFT: RLHF typically begins by fine-tuning a pre-trained LM with supervised learning on high-quality data for the downstream task(s) of interest (dialogue, summarization, etc.), to obtain a model π^{SFT}.

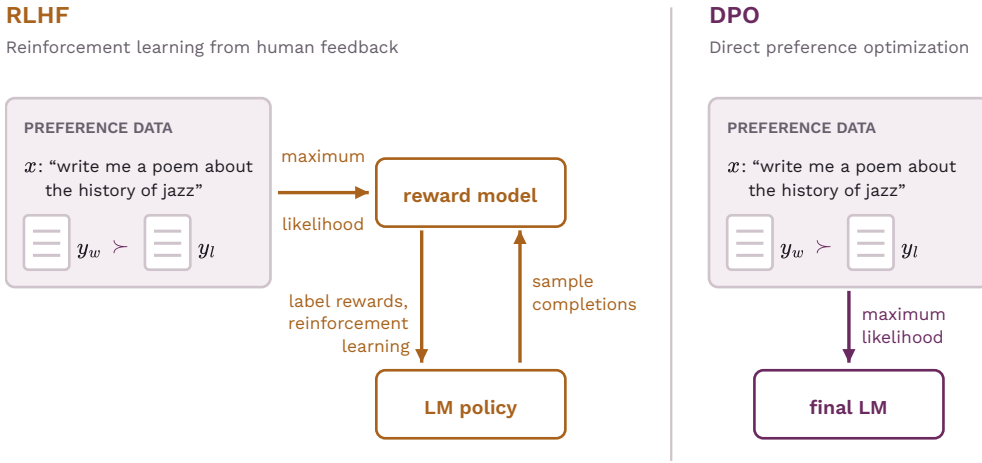


Figure 1. DPO optimizes for human preferences while avoiding reinforcement learning. Existing methods for fine-tuning language models with human feedback first fit a reward model to a dataset of prompts and human preferences over pairs of responses, and then use RL to find a policy that maximizes the learned reward. In contrast, DPO directly optimizes for the policy best satisfying the preferences with a simple classification objective, fitting an *implicit* reward model whose corresponding optimal policy can be extracted in closed form.

Redrawn from Figure 1 of Rafailov et al. (2023).

Reward Modelling Phase: In the second phase the SFT model is prompted with prompts x to produce pairs of answers $(y_1, y_2) \sim \pi^{\text{SFT}}(y \mid x)$. These are then presented to human labelers who express preferences for one answer, denoted as $y_w \succ y_l \mid x$ where y_w and y_l denotes the preferred and dispreferred completion amongst (y_1, y_2) respectively. The preferences are assumed to be generated by some latent reward model $r^*(y, x)$, which we do not have access to. There are a number of approaches used to model preferences, the Bradley-Terry (BT) model (Bradley and Terry, 1952) being a popular choice (although more general Plackett-Luce ranking models (Plackett, 1975; Luce, 2012) are also compatible with the framework if we have access to several ranked answers). The BT model stipulates that the human preference distribution p^* can be written as:

$$p^*(y_1 \succ y_2 \mid x) = \frac{\exp\left(r^*(x, y_1)\right)}{\exp\left(r^*(x, y_1)\right) + \exp\left(r^*(x, y_2)\right)}. \tag{1}$$

Assuming access to a static dataset of comparisons $\mathcal{D} = \left\{x^{(i)}, y_w^{(i)}, y_l^{(i)}\right\}_{i=1}^N$ sampled from p^*, we can parametrize a reward model $r_\phi(x, y)$ and estimate the parameters via maximum likelihood. Framing the problem as a binary classification we have the negative log-likelihood loss:

$$\mathcal{L}_R(r_\phi, \mathcal{D}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}}\left[\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))\right] \tag{2}$$

where σ is the logistic function. In the context of LMs, the network $r_\phi(x, y)$ is often initialized from the SFT model $\pi^{\text{SFT}}(y \mid x)$ with the addition of a linear layer on top of the final transformer layer that produces a single scalar prediction for the reward

value (Ziegler *et al.*, 2020). To ensure a reward function with lower variance, prior works normalize the rewards, such that $\mathbb{E}_{x,y\sim\mathcal{D}}\left[r_\phi(x,y)\right] = 0$ for all x.

RL Fine-Tuning Phase: During the RL phase, the learned reward function is used to provide feedback to the language model. Following prior works (Jaques *et al.*, 2017, 2020), the optimization is formulated as

$$\max_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi_\theta(y|x)}\left[r_\phi(x,y)\right] - \beta\mathbb{D}_{\mathrm{KL}}\left[\pi_\theta(y \mid x) \,||\, \pi_{\mathrm{ref}}(y \mid x)\right], \tag{3}$$

where β is a parameter controlling the deviation from the base reference policy π_{ref}, namely the initial SFT model π^{SFT}. In practice, the language model policy π_θ is also initialized to π^{SFT}.

4 Direct Preference Optimization

Motivated by the challenges of applying reinforcement learning algorithms on large-scale problems such as fine-tuning language models, our goal is to derive a simple approach for policy optimization using preferences directly. Unlike prior RLHF methods, which learn a reward and then optimize it via RL, our approach leverages a particular choice of reward model parameterization that enables extraction of its optimal policy in closed form, without an RL training loop. As we will describe next in detail, our key insight is to leverage an analytical mapping from reward functions to optimal policies, which enables us to transform a loss function over reward functions into a loss function over policies. This change-of-variables approach avoids fitting an explicit, standalone reward model, while still optimizing under existing models of human preferences, such as the Bradley-Terry model. In essence, the policy network represents both the language model and the (implicit) reward.

Deriving the DPO objective. We start with the same RL objective as prior work, Eq. (3), under a general reward function r. Following prior work (Peters and Schaal, 2007; Peng *et al.*, 2019; Korbak *et al.*, 2022; Go *et al.*, 2023), it is straightforward to show that the optimal solution to the KL-constrained reward maximization objective in Eq. (3) takes the form:

$$\pi_r(y \mid x) = \frac{1}{Z(x)}\pi_{\mathrm{ref}}(y \mid x)\exp\left(\frac{1}{\beta}r(x,y)\right), \tag{4}$$

where $Z(x) = \sum_y \pi_{\mathrm{ref}}(y \mid x)\exp\left(\frac{1}{\beta}r(x,y)\right)$ is the partition function. See Appendix A.1 for a complete derivation. Even if we use the MLE estimate r_ϕ of the ground-truth reward function r^*, it is still expensive to estimate the partition function $Z(x)$ (Korbak *et al.*, 2022; Go *et al.*, 2023), which makes this representation hard to utilize in practice. However, we can rearrange Eq. (4) to express the reward function in terms of its corresponding optimal policy π_r, the reference policy π_{ref}, and the unknown partition function $Z(\cdot)$. Specifically, we first take the logarithm of both sides of Eq. (4) and then with some algebra we obtain:

$$r(x,y) = \beta\log\frac{\pi_r(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta\log Z(x). \tag{5}$$

We can apply this reparameterization to the ground-truth reward r^* and corresponding optimal model π^*. Fortunately, the Bradley-Terry model depends only on the difference of rewards between two completions, i.e., $p^*(y_1 \succ y_2 \mid x) = \sigma(r^*(x, y_1) - r^*(x, y_2))$. Substituting the reparameterization in Eq. (5) for $r^*(x, y)$ into the preference model Eq. (1), the partition function cancels, and we can express the human preference probability in terms of only the optimal policy π^* and reference policy π_{ref}. Thus, the optimal RLHF policy π^* under the Bradley-Terry model satisfies the preference model:

$$p^*(y_1 \succ y_2 \mid x) = \frac{1}{1 + \exp\left(\beta \log \frac{\pi^*(y_2|x)}{\pi_{\text{ref}}(y_2|x)} - \beta \log \frac{\pi^*(y_1|x)}{\pi_{\text{ref}}(y_1|x)}\right)} \tag{6}$$

While Eq. (6) uses the Bradley-Terry model, we can similarly derive expressions under the more general Plackett-Luce models (Plackett, 1975; Luce, 2012).

Now that we have the probability of human preference data in terms of the optimal policy rather than the reward model, we can formulate a maximum likelihood objective for a parametrized policy π_θ. Analogous to the reward modeling approach (i.e. Eq. (2)), our policy objective becomes:

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]. \tag{7}$$

This way, we fit an implicit reward using an alternative parameterization, whose optimal policy is simply π_θ. Moreover, since our procedure is equivalent to fitting a reparametrized Bradley-Terry model, it enjoys certain theoretical properties, such as consistencies under suitable assumption of the preference data distribution (Bong and Rinaldo, 2022). In Section 5, we further discuss theoretical properties of DPO in relation to other works.

What does the DPO update do? For a mechanistic understanding of DPO, it is useful to analyze the gradient of the loss function $\mathcal{L}_{\text{DPO}}$. The gradient with respect to the parameters θ can be written as:

$$\nabla_\theta \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\beta \, \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \Bigg[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \Big[\underbrace{\nabla_\theta \log \pi(y_w \mid x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta \log \pi(y_l \mid x)}_{\text{decrease likelihood of } y_l} \Big] \Bigg],$$

where $\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$ is the reward implicitly defined by the language model π_θ and reference model π_{ref} (more in Section 5). Intuitively, the gradient of the loss function $\mathcal{L}_{\text{DPO}}$ increases the likelihood of the preferred completions y_w and decreases the likelihood of dispreferred completions y_l. Importantly, the examples are weighed by how much higher the implicit reward model $\hat{r}_\theta$ rates the dispreferred completions, scaled by β, i.e, how incorrectly the implicit reward model orders the completions, accounting for the strength of the KL constraint. Our experiments suggest the importance of this weighting, as a naïve version of this method without the weighting coefficient can cause the language model to degenerate.

5 Theoretical Analysis of DPO

5.1 Your Language Model Is Secretly a Reward Model

DPO is able to bypass both fitting an explicit reward and performing RL to learn the policy using a single maximum likelihood objective. Note the optimization objective Eq. (5) is equivalent to a Bradley-Terry model with a reward parameterization $r^*(x, y) = \beta \log \frac{\pi^*_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$ and we optimize our parametric model π_θ, equivalently to the reward model optimization in Eq. (2) under the change of variables. In this section we will build the theory behind this reparameterization, show that it does not constrain the class of learned reward models, and allows for the exact recovery of the optimal policy. We begin with by defining an equivalence relation between reward functions.

Definition 1. *We say that two reward functions $r(x, y)$ and $r'(x, y)$ are equivalent iff $r(x, y) - r'(x, y) = f(x)$ for some function f.*

It is easy to see that this is indeed an equivalence relation, which partitions the set of reward functions into classes. We can state the following two lemmas:

Lemma 1. *Under the Plackett-Luce, and in particular the Bradley-Terry, preference framework, two reward functions from the same class induce the same preference distribution.*

Lemma 2. *Two reward functions from the same equivalence class induce the same optimal policy under the constrained RL problem.*

The proofs are straightforward and we defer them to Appendix A.5. The first lemma is a well-known under-specification issue with the Plackett-Luce family of models (Plackett, 1975). The second lemma states that all reward functions from the same class yield the same optimal policy, hence for our final objective, we are only interested in recovering an arbitrary reward function from the optimal class.

Theorem 1. *Under mild assumptions, all reward classes consistent with the Plackett-Luce (and Bradley-Terry in particular) models can be represented with the reparameterization $r(x, y) = \beta \log \frac{\pi(y|x)}{\pi_{\text{ref}}(y|x)}$ for some model $\pi(y \mid x)$ and a given reference model $\pi_{\text{ref}}(y \mid x)$.*

Proof Sketch. Consider any reward function $r(x, y)$, which induces a corresponding optimal model $\pi_r(y \mid x)$, specified by Eq. (4). We will show that a reward function from the equivalence class of r can be represented using the reparameterization given above. We define the projection f as

$$f(r; \pi_{\text{ref}}, \beta)(x, y) = r(x, y) - \beta \log \sum_y \pi_{\text{ref}}(y \mid x) \exp\left(\frac{1}{\beta} r(x, y)\right) \tag{8}$$

The operator f simply normalizes the reward function with the logarithm of the partition function of π_r. Since the added normalization term is only a function of the prefix x, $f(r; \pi_{\text{ref}}, \beta)(x, y)$ is a reward function in the equivalence class of $r(x, y)$. Finally,

replacing r with the RHS of Eq. (5) (which holds for any reward function), we have $f(r; \pi_{\text{ref}}, \beta)(x, y) = \beta \log \frac{\pi_r(y|x)}{\pi_{\text{ref}}(y|x)}$. That is, the projection f produces a member of the equivalence class of r with the desired form, and we do not lose any generality in our reward model from the proposed reparameterization. □

6 Experiments

6.3 Generalization to a new input distribution

To further compare the performance of PPO and DPO under distribution shifts, we evaluate the PPO and DPO policies from our Reddit TL;DR summarization experiment on a different distribution, news articles in the test split of the CNN/DailyMail dataset (Nallapati *et al.*, 2016), using the best sampling temperatures from TL;DR (0 and 0.25). The results are presented in Table 1. For this new distribution, DPO continues to outperform the PPO policy by a significant margin.

Table 1. GPT-4 win rates vs. ground truth summaries for out-of-distribution CNN/DailyMail input articles.

	Win rate vs. ground truth	
Alg.	**Temp 0**	**Temp 0.25**
DPO	0.36	0.31
PPO	0.26	0.23

7 Discussion

Learning from preferences is a powerful, scalable framework for training capable, aligned language models. We have introduced DPO, a simple training paradigm for training language models from preferences without reinforcement learning. With virtually no tuning of hyperparameters, DPO performs similarly or better than existing RLHF algorithms, including those based on PPO; DPO thus meaningfully reduces the barrier to training more language models from human preferences.

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A Mathematical Derivations

A.1 Deriving the Optimum of the KL-Constrained Reward Maximization Objective

In this appendix, we will derive Eq. (4). Analogously to Eq. (3), we optimize the following objective:

$$\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi}\big[r(x,y)\big] - \beta\mathbb{D}_{\mathrm{KL}}\big[\pi(y|x)||\pi_{\mathrm{ref}}(y|x)\big] \tag{9}$$

under any reward function $r(x,y)$, reference model π_{ref} and a general non-parametric policy class. We now have:

$$
\begin{aligned}
&\max_{\pi} \mathbb{E}_{x\sim\mathcal{D},y\sim\pi}\big[r(x,y)\big] - \beta\mathbb{D}_{\mathrm{KL}}\big[\pi(y|x) \,||\, \pi_{\mathrm{ref}}(y|x)\big] \\
&\quad= \max_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi(y|x)}\left[r(x,y) - \beta\log\frac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)}\right] \\
&\quad= \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi(y|x)}\left[\log\frac{\pi(y|x)}{\pi_{\mathrm{ref}}(y|x)} - \frac{1}{\beta}r(x,y)\right] \\
&\quad= \min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{y\sim\pi(y|x)}\left[\log\frac{\pi(y|x)}{\frac{1}{Z(x)}\pi_{\mathrm{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right)} - \log Z(x)\right] \qquad (10)
\end{aligned}
$$

where we have partition function:

$$
Z(x) = \sum_{y} \pi_{\mathrm{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right).
$$

Note that the partition function is a function of only x and the reference policy π_{ref}, but does not depend on the policy π. We can now define

$$
\pi^*(y|x) = \frac{1}{Z(x)}\pi_{\mathrm{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right),
$$

which is a valid probability distribution as $\pi^*(y|x) \geq 0$ for all y and $\sum_y \pi^*(y|x) = 1$. Since $Z(x)$ is not a function of y, we can then re-organize the final objective in Eq. (10) as:

$$
\min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{E}_{y\sim\pi(y|x)}\left[\log\frac{\pi(y|x)}{\pi^*(y|x)}\right] - \log Z(x)\right] = \qquad (11)
$$

$$
\min_{\pi} \mathbb{E}_{x\sim\mathcal{D}}\left[\mathbb{D}_{\mathrm{KL}}(\pi(y|x) \,||\, \pi^*(y|x)) - \log Z(x)\right] \qquad (12)
$$

Now, since $Z(x)$ does not depend on π, the minimum is achieved by the policy that minimizes the first KL term. Gibbs' inequality tells us that the KL-divergence is minimized at 0 if and only if the two distributions are identical. Hence we have the optimal solution:

$$
\pi(y|x) = \pi^*(y|x) = \frac{1}{Z(x)}\pi_{\mathrm{ref}}(y|x)\exp\left(\frac{1}{\beta}r(x,y)\right) \qquad (13)
$$

for all $x \in \mathcal{D}$. This completes the derivation.

A.5 Proof of Lemma 1 and 2

> **Lemma 1 Restated.** *Under the Plackett-Luce preference framework, and in particular the Bradley-Terry framework, two reward functions from the same equivalence class induce the same preference distribution.*

Proof. We say that two reward functions $r(x,y)$ and $r'(x,y)$ are from the same equivalence class if $r'(x,y) = r(x,y) + f(x)$ for some function f. We consider the general Plackett-Luce (with the Bradley-Terry model a special case for $K = 2$) and denote the probability distribution over rankings induced by a particular reward function $r(x,y)$ as p_r. For any prompt x, answers $y_1, \ldots, y_K$ and ranking τ we have:

$$
\begin{aligned}
p_{r'}(\tau|y_1, \ldots, y_K, x) &= \prod_{k=1}^{K} \frac{\exp(r'(x, y_{\tau(k)}))}{\sum_{j=k}^{K} \exp(r'(x, y_{\tau(j)}))} \\
&= \prod_{k=1}^{K} \frac{\exp(r(x, y_{\tau(k)}) + f(x))}{\sum_{j=k}^{K} \exp(r(x, y_{\tau(j)}) + f(x))} \\
&= \prod_{k=1}^{K} \frac{\exp(f(x)) \exp(r(x, y_{\tau(k)}))}{\exp(f(x)) \sum_{j=k}^{K} \exp(r(x, y_{\tau(j)}))} \\
&= \prod_{k=1}^{K} \frac{\exp(r(x, y_{\tau(k)}))}{\sum_{j=k}^{K} \exp(r(x, y_{\tau(j)}))} \\
&= p_r(\tau|y_1, \ldots, y_K, x),
\end{aligned}
$$

which completes the proof. □

Abridged and re-set for the Postext Cookbook from arXiv:2305.18290v3 (CC BY 4.0, https://creativecommons.org/licenses/by/4.0/). Sections 2 and 5.2, most of section 6 and appendices A.2–A.4, A.6 and B–E are cut; section numbers are the original ones, equations are renumbered. Figure 1 is redrawn in code. Set in Spectral, Work Sans and JetBrains Mono (SIL OFL); formulas by MathJax.